

STEP 1: Columns You Should Delete

Column	Reason for Deleting (Short)
player_id	Unique identifier, not useful for prediction.
first_name	Player identity does not contribute to prediction.
last_name	Player identity does not contribute to prediction.
name_x	Duplicate player name after merging.
last_season_x	Historical metadata, not required.
current_club_id	Identifier column. Current club name is retained.
player_code	Unique identifier with no analytical value.
country_of_birth	Birthplace is less relevant than citizenship.
city_of_birth	Birth location has no impact on prediction.
agent_name	Agent information is outside the project scope.
image_url	Image metadata.
current_national_team_id	Identifier already represented through international statistics.
url_x	URL metadata.
current_value	Duplicate of market_value_in_eur.
highest_value	Duplicate of highest_market_value_in_eur.
average_value	Derived value that can be recreated if required.
min_value	Low analytical importance.
club_id	Identifier column.
club_code	Club identifier.
name_y	Duplicate club name after merging.
domestic_competition_id	Competition code already retained.
total_market_value	Contains 100% missing values.
coach_name	Coach information is not related to player prediction.
last_season_y	Duplicate metadata column.
filename	File metadata.
url_y	URL metadata.
competition_id	Identifier column.
country_id	Country name is already available.
url	URL metadata.

STEP 2: Cleaned Dataset (Columns to Keep)

Player Profile

Column	Why Kept
country_of_citizenship	Represents the player's nationality.
date_of_birth	Used to derive the player's age in later stages.
sub_position	Specifies the player's detailed playing role.
position	Indicates the primary playing position.
foot	Represents the player's preferred foot.
height_in_cm	Important physical attribute.
contract_expiration_date	Useful career-related information.

International Performance

Column	Why Kept
international_caps	Measures international experience.
international_goals	Measures international performance.

Club Information

Column	Why Kept
current_club_domestic_competition_id	Represents the player's current league.
current_club_name	Identifies the player's current club.

Market Value

Column	Why Kept
market_value_in_eur	Current market value of the player.
highest_market_value_in_eur	Highest recorded market value.
value_growth	Indicates the player's market value growth.

Match Performance

Column	Why Kept
matches_played	Indicates overall playing experience.
total_minutes	Represents total playing time.
avg_minutes	Average minutes played per match.
total_goals	Measures scoring ability.
avg_goals	Average goals scored per match.
total_assists	Measures playmaking ability.
avg_assists	Average assists per match.
yellow_cards	Indicates player discipline.
red_cards	Indicates player discipline.

Transfer Statistics

Column	Why Kept
transfer_count	Number of career transfers.
total_transfer_fee	Total transfer fee throughout the career.
highest_transfer_fee	Highest transfer fee received.

Club Statistics

Column	Why Kept
squad_size	Represents the club's squad size.
average_age	Average age of players in the club.
foreigners_number	Number of foreign players in the club.
foreigners_percentage	Percentage of foreign players.
national_team_players	Number of international players in the club.

stadium_name	Represents the club's home stadium.
stadium_seats	Stadium seating capacity.
net_transfer_record	Indicates the club's transfer spending pattern.

Competition Information

Column	Why Kept
competition_code	Unique competition identifier.
name	Competition name.
sub_type	Competition sub-category.
type	Competition type.
country_name	Country where the competition is held.
domestic_league_code	Domestic league identifier.
confederation	Football governing confederation.
total_clubs	Number of clubs participating in the competition.

Target Variables

Column	Why Kept
Player_Future_Potential_Score	Numerical target variable ranging from 0–100 used for prediction.
Player_Future_Potential_Rating	Categorical target variable (Outperforming, Average, Underperforming).

STEP 3: Analysis of Cleaned Dataset Columns

Player Profile

These columns describe the player's personal and physical characteristics, including nationality, playing position, preferred foot, height, and contract information. They provide the basic attributes required for evaluating future player potential.

International Performance

These columns represent the player's experience and achievements at the international level. Greater international exposure often indicates higher experience and competitive maturity.

Club Information

These columns identify the player's current club and domestic competition, providing context about the competitive environment in which the player performs.

Market Value

Market value columns represent the financial worth and growth of a player. These variables reflect demand, performance, and career progression, making them strong indicators of future potential.

Match Performance

These columns summarize the player's on-field performance through appearances, playing time, goals, assists, and disciplinary records. They are the primary indicators of football performance.

Transfer Statistics

Transfer-related columns describe a player's transfer history and market demand. They help analyze career progression and transfer value.

Club Statistics

These columns provide information about the player's club environment, including squad characteristics, international representation, and transfer activity.

Competition Information

Competition-related columns describe the league or tournament in which the player competes. The level of competition influences player development and overall performance.

Target Variables

- **Player_Future_Potential_Score** is the numerical target variable ranging from **0 to 100**, representing the estimated future potential of a football player.
- **Player_Future_Potential_Rating** categorizes the numerical score into three classes:
 - Outperforming
 - Average
 - Underperforming

These target variables will be used to train and evaluate the AI model.

STEP 3: Final Dataset Summary

After cleaning:

- **Player Profile:** 7 columns
- **International Performance:** 2 columns
- **Club Information:** 2 columns
- **Market Value:** 3 columns
- **Match Performance:** 9 columns
- **Transfer Statistics:** 3 columns
- **Club Statistics:** 8 columns
- **Competition Information:** 8 columns
- **Target Variables:** 2 columns

Total Rows Before Cleaning: 50,149

Rows After Cleaning: 50,148

Total Columns Before Cleaning: 73

Columns Removed: 29

Final Columns After Cleaning: 44

Numerical Columns: 26

Categorical/Text Columns: 18

Target Columns: 2